The universal series has exact contact at sufficiently large primes

$r = 2,\ p = 7$
$p > 3r - 1$
$v_p(H_r(t)) = 3e$

$r = 3,\ p = 11$
$p > 3r - 1$
$v_p(H_r(t)) = 6e$

$r = 4,\ p = 13$
$p > 3r - 1$
$v_p(H_r(t)) = 10e$

$v_p(t) = e \geq 1$
Unique initial coefficient is a p-adic unit